Alert

Unit ID: **534.LQ145 - Girth Gear** Asset ID: **52305241**

Description: **Cement Mill #4 Gearbox Lubrication Unit**

Account Information

ID: **462031**
Name: **ASEC - ACC**
Address: **, , EG**

Sample Information

Sample ID: **26124405254**
Service Level: **Enhanced**
Bottle ID: **b098085823**
Tested Lubricant: **MOBILGEAR 600 XP 460**

Equipment Information

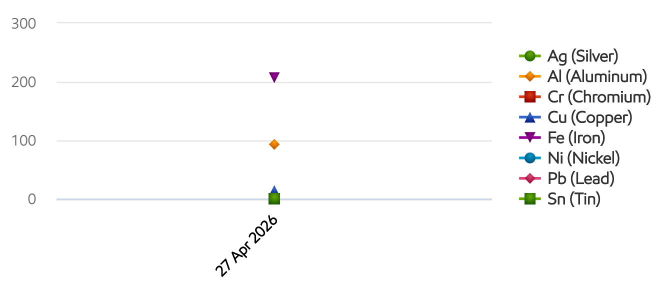
Asset Class: **Circulating System**
Manufacturer: **Flender**
Model:
Lubricant: **MOBILGEAR 600 XP 460**

Sample Data & Trends

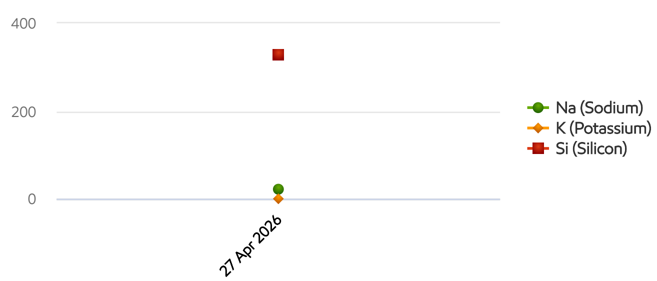
Sample Info	Report Status	Alert
	Sample ID	26124405254
	Service Level	Enhanced
	Bottle ID	b098085823
	Tested Lubricant	MOBILGEAR 600 XP 460
	Sampled	27 Apr 2026
	Reported	05 May 2026
	Equipment Age	
	Oil Age	
	Make-up Volume	
	Oil Changed	
	Filter Changed	
Lubricant	Contamination Rating	Alert
	Equipment Rating	Alert
	Lubricant Rating	Normal
	ISO Code (4/6/14)	
	Particle Count >4um	
	Particle Count >6um	
	Particle Count>14um	
	PQ Index	194
	Visc@40C (cSt)	447.4
	Oxidation (Ab/cm)	0
	Water (Vol%)	0.300
Wear (ppm)	Ag (Silver)	0
	Al (Aluminum)	94
	Cr (Chromium)	0
	Cu (Copper)	13
	Fe (Iron)	207
	Mo (Molybdenum)	0
	Ni (Nickel)	0
	Pb (Lead)	0
	Sn (Tin)	0
Contaminant (ppm)	K (Potassium)	0
	Na (Sodium)	22
	Si (Silicon)	330
Additive (ppm)	B (Boron)	103
	Ba (Barium)	0
	Ca (Calcium)	1451
	Mg (Magnesium)	67
	P (Phosphorus)	427
	Zn (Zinc)	7

Viscosity

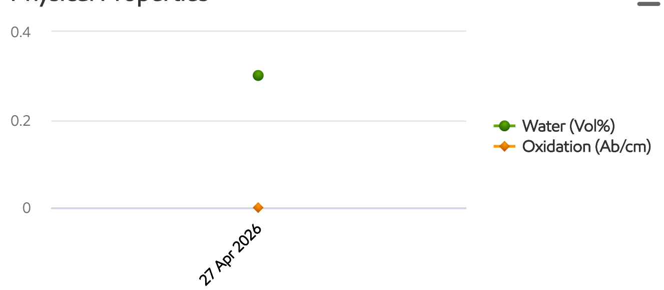
Wear



Contaminants



Physical Properties



Mobil™

Alert

Unit ID: **534.LQ145 - Girth Gear** Asset ID: **52305241**

Description: **Cement Mill #4 Gearbox Lubrication Unit**

Account Information	Sample Information	Equipment Information
ID: **462031**	Sample ID: **26124405254**	Asset Class: **Circulating System**
Name: **ASEC - ACC**	Service Level: **Enhanced**	Manufacturer: **Flender**
Address: **, , EG**	Bottle ID: **b098085823**	Model:
	Tested Lubricant: **MOBILGEAR 600 XP 460**	Lubricant: **MOBILGEAR 600 XP 460**

Recommendation/Comments

ACTION REQUIRED - ELEVATED WEAR METALS LEVELS. Determine cause of elevated wear metals levels and take corrective action. If the wear metals levels are elevated for the first time with no history of an increasing trend, then resample immediately to confirm condition. Because silicon levels are not also elevated, abrasive dirt is not the cause of this condition. Possible causes of elevated wear metals include: a. Worn of oil-wetted parts; b. Contamination from previous fill; c. Over-extended oil service life; d. Normal break-in of new equipment. Continue to closely monitor the equipment and oil for signs of further condition regressions. Contact your ExxonMobil representative for further assistance if necessary.

ACTION REQUIRED - ELEVATED IRON LEVEL. Determine source of Iron (Fe) and take corrective action. If the iron level is elevated for the first time with no history of an increasing trend, then resample immediately to confirm condition. Possible sources of iron include: a. Machinery component wear; b. Metal-to-metal contact; c. Corrosion; d. Abrasive dirt contamination. The oil may be unfit for continued use and should be closely monitored for further condition regressions. Inspect the equipment for signs of damage. Change the oil after corrective measures have been implemented. Contact your ExxonMobil representative for further assistance if necessary.

ACTION REQUIRED - EXCESSIVE FERROUS DEBRIS. Determine cause of ferrous debris and take corrective action. If the ferrous debris level is elevated for the first time with no history of an increasing trend, then resample immediately to confirm condition. Potential causes of excessive ferrous debris include: a. Abnormal metallic component wear caused by abrasive dirt (or other contaminants); b. Material fatigue from heavy loads or other adverse operating conditions; c. Corrosive wear due to water, acids, or other contaminants in the lubricant; d. Normal "break-in" wear of new or recently replaced equipment. Inspect the equipment for signs of damage. Change the oil after corrective measures have been implemented. Contact your ExxonMobil representative for further assistance if necessary.

ACTION REQUIRED - ELEVATED SILICON LEVEL. Determine source of Silicon (Si) and take corrective action. If the silicon level is elevated for the first time with no history of an increasing trend, then resample immediately to confirm condition. If the wear metals levels are low, then the silicon likely exists in a non-abrasive form. Possible sources of non-abrasive silicon include: a. Defoamants; b. Sealant compounds. If wear metals levels are elevated, then the silicon likely exists in an abrasive form. Possible sources of abrasive silicon include: a. Abrasive dirt; b. Contamination from environment during an equipment overhaul or shutdown. Other possible sources of silicon include: a. Inefficient or leaking air intake filters; b. Leaking pipes; c. Clogged oil bath air filter; d. Poor oil filtration; e. Dirty sample containers or improper sampling procedures. In landfill gas applications, silicon can enter the engine as a siloxane. This material is a gas and is not normally abrasive; however, siloxanes can easily mask the presence dirt. The oil may be unfit for continued use and should be monitored closely for further condition regressions. Prolonged usage of lubricants with elevated levels of abrasive silicon can cause severe wear and damage to the equipment. Inspect the equipment for signs of excessive wear and change the oil immediately if damage is detected. Contact your ExxonMobil representative for further assistance if necessary.

ACTION REQUIRED - EXCESSIVE WATER CONTAMINATION. Determine source of water and take corrective action. Analyze the oil with an on-site water test if accessible. Potential sources of water include: a. Leaking seals; b. Condensation resulting from low operating temperatures or prolonged shutdowns; c. Compromised oil cooler; d. Direct exposure to atmosphere as a result of compromises to system components (i.e. improperly sealed tank fill access point. Remove water from oil using centrifuges or dehydration units, or drain the oil and recharge the system with new oil. Resample the system after corrective measures have been implemented. Contact your ExxonMobil representative for further assistance if necessary.

ACTION REQUIRED - UNSATISFACTORY OIL CONDITION. Test results indicate that the condition of this oil sample is unsatisfactory. The oil may be unfit for continued use and should be closely monitored further condition regressions. Contact your ExxonMobil representative for further assistance if necessary.

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	Tested Lubricant: **MOBILGEAR 600 XP 460**	Lubricant: **MOBILGEAR 600 XP 460**

ADMINISTRATION - PARTICLE COUNT DATA NOT REPORTED. The sample is heavily contaminated with visible sediment. Oil samples that contain significant levels of any contaminant can damage the laboratory's precision equipment. As a result, the particle count test was not completed, and analysis could not be provided. Before sending future samples to the laboratory, please ensure that no signs of visible sediment contamination are present. Consider the following: 1. The sample may not be representative of the system if it was acquired from a non-turbulent area (i.e. tank bottom); 2. Installing sample ports helps ensure that the sample is taken from the same location every time; 3. Tank fill caps and breathers might not be properly secured; 4. The filter / centrifuge may be operating incorrectly. If the sample was representative of the system, change the oil immediately to prevent equipment damage. Contact your ExxonMobil representative for further assistance if necessary.

Sample Timeline

- 27 Apr 2026 12:38 PM - Abd El Rahman Khaled - In Service Oil Sample : Comments: Photo: